The Problem

Content management in health care governs the written artifacts that determine how care is authorized, delivered, and paid for. Unlike document management, which stores and retrieves files, content management treats the meaning of the text as the managed asset. Every paid or denied claim is ultimately the execution of prose someone first had to interpret. The core technical problem naturally is translation: the summarizing of policy content, comparing changes between versions, and converting written policy into rule trees or code.

CMS's National Coverage Determinations bind every Medicare contractor, and Local Coverage Determinations fill the gaps where none exists. These NCDs can take up to nine months to issue, so the rules never fully settle (CMS, n.d.-a, n.d.-b). Commercial payers layer on their own policies, and someone still has to re-encode all of it by hand each time something changes. The cost is substantial. Physician-practice interactions with health plans run $23–31 billion nationally per year (Casalino et al., 2009), $2,161–$3,430 per physician annually in New York and Pennsylvania alone (Morley et al., 2013). More than one in four physicians report prior authorization has led to a serious adverse patient event (AMA, 2025).

The regulatory push toward fixing this is already underway. CMS-9115-F requires FHIR APIs for patient access and payer-to-payer exchange (CMS, 2020); CMS-0057-F extends that to prior authorization itself (CMS, 2024), with a Provider Access API and Prior Authorization API due by January 1, 2027. The standards exist, so encoding the rules is the bottleneck.

Recent NLP work offers a path through that bottleneck. Self-Refine, where a model drafts, critiques, and revises its own output, yields ~20% average gains with no retraining (Madaan et al., 2023). Reflexion, which stores an agent's self-critiques as memory across attempts, pushed HumanEval pass@1 to 91% (Shinn et al., 2023). RAG grounds output in cited source text rather than parametric memory (Ayala & Bechard, 2024), and JSON Logic offers a minimal, portable execution target (JsonLogic, n.d.). Combined, these four pieces make compiling a policy PDF into an auditable, self-verifying, executable artifact a realistic product, not a wishful one.

Opportunity and Risk

Cotiviti's has the opening to improve the slow, manual process of onboarding new insurance policies. Subject matter experts (SMEs) and rules analysts must re-read prose and rewrite code every time a policy shifts. Instead, one agent could draft a rule. A second could try to break it with adversarial edge-case claims. A third could patch it until the two stop disagreeing. That gets a first-draft rule out in seconds instead of days, and it shifts the SME into a review role instead of a typing one. Diffing two compiled artifacts instead of two PDFs surfaces behavioral differences rather than editorial ones. This fits naturally with the quarterly NCD/LCD cycle and the annual CPT/ICD refresh. Retrieval citations tie each clause back to a specific span of source text, which is exactly the paper trail an appeal or compliance review needs. And Cotiviti's 2025 acquisition of Edifecs already gives it the FHIR plumbing to expose all of this through the APIs CMS-0057-F now requires.

The risks must not be overlooked, however. RAG reduces hallucination, but it doesn't eliminate it (Ayala & Bechard, 2024), and a wrong rule shipped at scale means claims processed incorrectly at scale too. In June 2026, the AMA House of Delegates adopted a new policy that specifically calls for "regulating AI use" in prior authorization and coverage decisions (AMA, 2026), so shipping a black box here risks turning Cotiviti into a case of algorithmic denial. Optum and Availity are already circling this space, and speed to a credible product will likely matter more than piling on features. There's a subtler failure mode as well. An aggressive self-critique loop can quietly import clinical knowledge that the policy text never actually states, producing a rule that looks correct but has drifted from its source. That's harder to catch than an outright hallucination, and it needs to be designed against from day one.

Cotiviti's Potential Next Steps

Cotiviti should aim to start inside Payment Policy Management, where the compile-and-critique loop should be positioned as a multiplier for the existing SME and rules-analyst team rather than a replacement for it. Success here is best measured in SME hours saved per policy encoded, not in the number of decisions the system makes autonomously. This framing matters as much internally for adoption as it does externally, especially given the AMA's newly adopted posture toward regulating AI in coverage decisions (AMA, 2026).

Once the pipeline is producing compiled, self-verified rules efficiently, the diff artifact it generates along the way becomes a product in its own right rather than a mere byproduct of the compilation process. Piped through the FHIR infrastructure Cotiviti inherited from its 2025 acquisition of Edifecs, that artifact could be exposed as a policy-change feed. It can become a CMS-aligned API that providers and downstream payers could subscribe to directly, turning what would otherwise be an internal back-office tool into something marketable.

Longer term, Cotiviti should consider publishing an open benchmark: a set of CMS NCD/LCD policy pairs annotated with SME-graded expected rule deltas. This both invites scrutiny of its own approach and raises the bar competitors must clear. Given that Optum and Availity are already circling this same space, an openly published, independently verifiable evaluation set would make it considerably harder for a competitor to win the narrative with a cherry-picked demo, and it would let Cotiviti answer AMA-style scrutiny with evidence rather than a press release.

The regulatory push, the cost pressure, and the maturity of the underlying AI tooling have all converged within the same two-year window. The real question facing Cotiviti, then, isn't whether large language models will be used to encode policy. That's already a reality somewhere, officially or otherwise. It's whether the pipeline that ends up doing it ships with self-verification and traceable citations built in from the outset, rather than bolted on only after the first bad rule makes the news.

References

American Medical Association. (2025). 2025 AMA prior authorization physician survey. https://www.ama-assn.org/system/files/prior-authorization-survey.pdf

American Medical Association. (2026, June 9). AMA adds more to its game plan to fix prior authorization. https://www.ama-assn.org/practice-management/prior-authorization/ama-adds-more-its-game-plan-fix-prior-authorization

Ayala, O., & Bechard, P. (2024). Reducing hallucination in structured outputs via retrieval-augmented generation. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track) (pp. 228–238). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.naacl-industry.19

Casalino, L. P., Nicholson, S., Gans, D. N., Hammons, T., Morra, D., Karrison, T., & Levinson, W. (2009). What does it cost physician practices to interact with health insurance plans? Health Affairs, 28(4), w533–w543. https://doi.org/10.1377/hlthaff.28.4.w533

Centers for Medicare & Medicaid Services. (2020, April 21). Interoperability and Patient Access fact sheet. https://www.cms.gov/newsroom/fact-sheets/interoperability-and-patient-access-fact-sheet

Centers for Medicare & Medicaid Services. (2024, January 17). CMS interoperability and prior authorization final rule (CMS-0057-F) [Fact sheet]. https://www.cms.gov/newsroom/fact-sheets/cms-interoperability-and-prior-authorization-final-rule-cms-0057-f

Centers for Medicare & Medicaid Services. (n.d.-a). Medicare coverage determination process. https://www.cms.gov/medicare/coverage/determination-process

Centers for Medicare & Medicaid Services. (n.d.-b). National coverage determination process & timeline. https://www.cms.gov/cms-guide-medical-technology-companies-and-other-interested-parties/coverage/national-coverage-determination-process-timeline

JsonLogic. (n.d.). JsonLogic: Build complex rules, serialize them as JSON, and execute them in any language. https://jsonlogic.com/

Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegreffe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., Welleck, S., Majumder, B. P., Gupta, S., Yazdanbakhsh, A., & Clark, P. (2023). Self-Refine: Iterative refinement with self-feedback (arXiv:2303.17651). https://arxiv.org/abs/2303.17651

Morley, C. P., Badolato, D. J., Hickner, J., & Epling, J. W. (2013). The impact of prior authorization requirements on primary care physicians' offices: Report of two parallel network studies. Journal of the American Board of Family Medicine, 26(1), 93–95. https://doi.org/10.3122/jabfm.2013.01.120062

Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning (arXiv:2303.11366). https://arxiv.org/abs/2303.11366